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Design and Implementation of a Solar-Diesel Hybrid System for Oil and Gas Rigs

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Abstract

Remote oil and gas (O&G) rigs primarily depend on diesel generators (DGs), which results in high fuel costs, significant carbon dioxide (CO₂) emissions, and logistical challenges. This paper presents the design, deployment, and performance assessment of a solar-diesel hybrid system implemented at a land rig in Saudi Arabia. The hybrid system consists of solar PV arrays, battery storage, and diesel generators managed by a smart Energy Management System (EMS). The study is divided into two phases: the first targets non-critical loads such as camp and lighting systems; the second explores a solar-battery-diesel solution to support drilling operations. Methodologically, this project involved extensive load profiling, site-specific solar irradiance analysis, hybrid system simulation, and phased implementation.

The solar-diesel hybrid design for the camp included rooftop and portable solar PV panels totaling 390 kW, DC/AC inverters, and EMS integration. For lighting, solar-powered portable tower and crown lights were deployed. Results demonstrated a reduction of 207,676 liters of diesel and 557 tons of CO₂ annually, alongside operational savings of over \$17,000 USD. The hybrid drilling operations system, still in the planning phase, is projected to save 1.2 million liters of diesel annually and reduce emissions by 3,180 tons. Findings indicate significant environmental and economic benefits, validating the system's scalability across other rigs in the region.

This work supports Saudi Arabia's Vision 2030 and global energy transition efforts, demonstrating the feasibility and advantages of integrating renewables into critical O&G operations.

Introduction

Since the 1950s, O&G have driven industrial growth, relying on remote onshore and offshore rigs powered by DGs. While DGs ensure reliable operations, they contribute significantly to CO₂ emissions, worsening climate change. In 2023, 1,577 O&G rigs operated globally, with 294 active rigs in Saudi Arabia as of 2024. To address this, Saudi Arabia aims to achieve net-zero emissions by 2060, with a 2030 interim goal of reducing CO₂ emissions by 278 million tons under Vision 2030 [1].

This study examines a Solar-Diesel hybrid system implemented at a land rig, integrating solar PV panels with DGs to reduce diesel use, costs, and emissions. Managed by a smart Energy Management System (EMS), the project demonstrates a scalable energy model, with plans for a solar-battery-diesel system to

further support sustainability goals. These efforts highlight the potential for renewable energy solutions to transform the O&G sector, aligning operational efficiency with environmental responsibility and supporting global climate objectives.

The motivation

Rig count; worldwide and Saudi Arabia

As of 2024, the global count of O&G rigs stands at 1,577, with 1,337 onshore rigs and 240 offshore rigs. During periods of elevated oil prices, such as in 2014, the global rig count exceeded 3,400 due to increased exploration and drilling activity.

Similarly, in Saudi Arabia (KSA), 294 rigs were operational in 2024, with 265 onshore and 29 offshore rigs, highlighting the Kingdom's heavy reliance on onshore operations for its O&G activities.

The majority of these rigs are powered by DGs, given their remote locations and lack of access to grid electricity. This reliance on diesel results in substantial fuel consumption, leading to high operational costs, logistical challenges, and considerable CO₂ emissions. These environmental concerns emphasize the critical need for more sustainable and cost-effective energy solutions in the O&G sector.

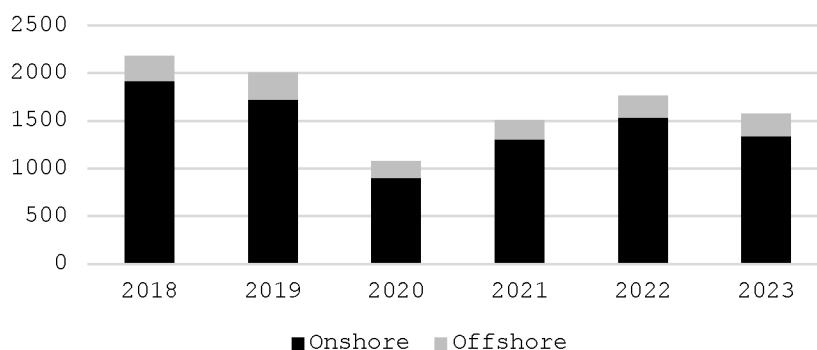


Figure 1—Worldwide rig count 2018-2023

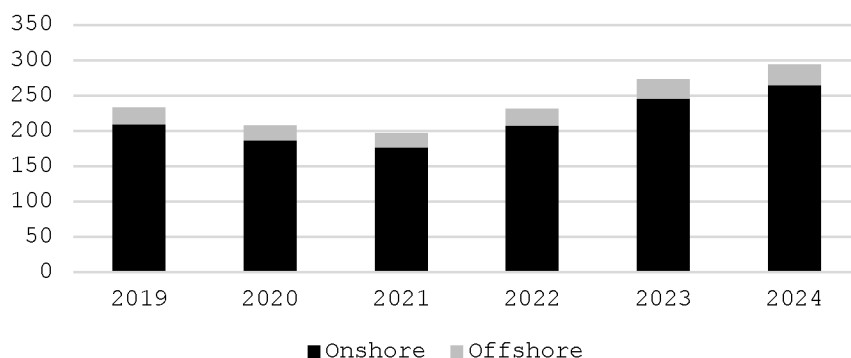


Figure 2—Saudi Arabia rig count 2019-2024

The candidate rig diesel consumption

This project represents a pioneering initiative in integrating hybrid energy systems into O&G operations. The project is divided into two parts. The first part involves the design and implementation of a Solar-Diesel hybrid system for the rig camp and lighting, which handles non-critical loads, utilizing the existing DGs. The second part focuses on evaluating a Solar-Diesel-Battery hybrid system for the rig's drilling operations. Upon successful implementation of the first phase, the project will be scaled to additional O&G rigs, serving as a foundation and motivation for adopting hybrid energy solutions for drilling operations in the future.

The energy consumption is categorized into four primary areas, forming the foundation for the phased implementation of the hybrid system. The first phase targets the camp, tower lights, and crown lights, while the second phase focuses on drilling operations due to their critical power demands.

The rig camp includes essential facilities such as accommodation, catering services, and water heating systems for personnel. This section is powered by a dedicated diesel generator with a capacity of 568 kVA (455 kW), which typically operates near its full load. As a result, the monthly diesel consumption for the camp is approximately 16,500 liters, underscoring the high energy demand of these support functions.

Illumination of the rig site is provided by eight portable tower lights, all powered by diesel generators. These lights consume a total of 5,550 liters of diesel each month, highlighting the substantial energy required to ensure operational safety and visibility, particularly during night shifts and adverse weather conditions.

The rig tower itself is equipped with four crown-mounted lighting systems to provide elevated site illumination. The setup includes 23 lights rated at 400 W each and 9 lights rated at 150 W each, with a total power demand of 10.55 kW. These lights are also powered by diesel generators, contributing an additional 2,550 liters to the monthly diesel consumption.

Drilling operations represent the most energy-intensive aspect of the rig's activities. The rig operates four diesel generators, each rated at 1,750 kVA (1,200 kW). Under normal operating conditions, three of these generators run at approximately 80% load, supplying around 2,880 kW of power. During peak scenarios, such as stuck pipe events, all four generators are required, pushing the load to approximately 4,320 kW (or roughly 4 MW). Monthly diesel consumption for drilling operations reaches around 250,000 liters, emphasizing the scale of fuel demand and the critical need for more efficient and sustainable power solutions.

The rig consumes 24,600 liters of diesel monthly for the camp and lighting systems, while drilling operations use approximately 250,000 liters. This highlights the substantial fuel demands and costs typical of onshore O&G rigs, emphasizing the need for sustainable energy solutions.

The rig diesel cost

As of October 2024, Saudi Aramco has set the diesel price in Saudi Arabia at 1.15 SAR (0.31 USD) per liter, significantly lower than global averages due to the Kingdom's role as a major oil producer. However, these costs still impose a substantial burden on continuous operations. For the candidate rig, with a monthly consumption of 24,600 liters for the camp and lighting and 250,000 liters for drilling operations, the costs amount to 28,290 SAR (7,544 USD) for the camp and 287,500 SAR (76,667 USD) for drilling. Annually, this equates to 339,480 SAR (90,528 USD) for the camp and 3,450,000 SAR (920,000 USD) for drilling [2].

Extrapolating these figures to the 273 rigs in Saudi Arabia (2023), the annual diesel cost for rig camps is approximately 93 million SAR (24.8 million USD), while for drilling operations, it reaches 942 million SAR (251 million USD).

Globally, diesel prices in the O&G sector average 1.18 USD per liter, significantly impacting operational costs. Assuming rigs similar to the candidate rig, with monthly consumption of 24,600 liters for camps and lighting and 250,000 liters for drilling, the monthly costs are approximately 28,910 USD for camps and 295,000 USD for drilling. Scaling these costs to the 1,577 rigs worldwide (2023), the annual diesel expenditure amounts to 547 million USD for camps and 5.6 billion USD for drilling operations. These figures underscore the immense financial impact of diesel consumption in the O&G sector and the need for cost-effective, sustainable alternatives.

The rig CO₂ emissions

The diesel consumption for the rig camp and lighting systems, at 24,600 liters per month, results in significant CO₂ emissions. Using a standardized CO₂ emissions factor of 2.68 kilograms of CO₂ per liter of diesel, this equates to approximately 65,928 kilograms (65.9 tons) of CO₂ each month or 791 tons annually.

for camp operations. For drilling operations, which consume 250,000 liters of diesel monthly, the emissions amount to 670,000 kilograms (670 tons) per month or 8,040 tons annually [3].

Saudi Arabia, with 273 rigs operational in 2023, reflects a substantial environmental impact if each rig exhibits similar diesel consumption patterns as the candidate rig. For camp operations alone, the total annual CO₂ emissions are approximately 216,183 tons, while drilling operations contribute an additional 2.2 million tons annually. This underscores the scale of emissions associated with diesel use across the Kingdom's O&G sector.

On a global scale, with 1,577 rigs active in 2023, the emissions from camp operations are estimated at 1.25 million tons annually, and drilling operations contribute approximately 12.7 million tons annually. These figures highlight the significant global environmental footprint of diesel-powered O&G rigs, emphasizing the pressing need for alternative energy solutions to mitigate CO₂ emissions and their associated climate impacts.

Solar-diesel hybrid system design for the rig camp and lighting

The design of the Solar-Diesel hybrid system for the rig camp and lighting aims to minimize diesel consumption and reduce CO₂ emissions while ensuring reliable power generation. The following provides an overview of the system's core components and design approach.

Data gathering

This section outlines the data collection process necessary for designing the proposed hybrid system, focusing on three primary load categories: the main camp, tower lights, and crown lights.

The main camp is powered by a diesel generator that typically operates near full capacity to meet the substantial energy demand of the accommodation and support facilities. The generator, with a nameplate capacity of 455 kW, consumes approximately 326.6 MWh of energy per month when operating at full load. During a site visit, operational measurements were recorded, including a current of 1,181 A, voltage of 220 V, and a power factor of 0.835. These values correspond to an actual power output of approximately 449.5 kW, which translates to a monthly energy consumption of 323.64 MWh. This measured output closely aligns with the theoretical load, confirming the high energy utilization at the camp.

Lighting across the rig site is provided by eight portable tower lights, which serve to ensure safe working conditions for crew members during nighttime and low-visibility conditions. According to the engineering team, each tower light consumes around 650 liters of diesel per month, leading to a total monthly diesel consumption of approximately 5,200 liters. This represents a significant portion of the lighting energy demand for the site.

Additional lighting is provided by 32 crown lights mounted on the rig tower, with eight lights positioned on each side to deliver comprehensive overhead illumination. These lights are specifically designed to enhance visibility across the operational area from an elevated position. Each side of the tower consumes about 748 liters of diesel monthly, which results in a combined total of approximately 2,992 liters per month, or 35,904 liters annually, for all 32 crown lights.

A summary of the energy consumption, diesel usage, cost implications, and carbon emissions associated with the camp and lighting systems for the candidate rig is provided in the following table. This data forms the foundation for optimizing system sizing and evaluating the potential benefits of transitioning to a solar-diesel hybrid configuration.

Table 1—Energy, diesel consumption, cost, and CO2 emissions of the candidate camp and lighting

	Main Camp	Tower Lights	Crown Lights	Total
Quantity (each)	1	8	32	N/A
Power Rating (kW)	455	10.26	6.68	471.94
Annual Energy Consumption (MWh)	3,931.2	90	59	4,079.57
Annual Diesel Consumption (L)	295,200	62,400	35,904	393,504
Annual Diesel Cost (USD)	90,528	19,136	11,011	120,675
Annual CO2 Emissions (tons)	791.2	167	96	1,055

System design based on the gatehred data

The Solar-Diesel hybrid system designed for deployment at the candidate rig's camp and lighting systems is structured to address three critical power consumption areas. Foremost among these is the main camp, which represents the largest load within the camp and lighting segment. To offset this demand, the system incorporates three portable solar generators, each rated at 70 kW, providing a combined capacity of 210 kW. Operating under an average of seven peak sun hours per day, this configuration is expected to generate approximately 1,470 kWh daily, amounting to 536.55 MWh annually. In addition to these mobile units, photovoltaic (PV) modules will be installed on the rooftops of 40 caravans. Each caravan is designed to support 4.5 kW of PV capacity, adding another 180 kW to the overall solar contribution.

Given that the solar PV system produces direct current (DC) while the camp requires alternating current (AC), the system integration necessitates reliable DC/AC conversion. To achieve this, the HUAWEI SUN-2000-50KTL-M0 inverter has been selected. This model offers a 50 kW AC output along with Maximum Power Point Tracking (MPPT) capabilities to ensure efficient energy conversion. The total 390 kW DC input—comprising 210 kW from the portable solar generators and 180 kW from the rooftop modules—will be managed by four inverters dedicated to the portable units and three inverters assigned to the caravan-mounted PV systems. This configuration ensures stable and efficient power delivery to meet the camp's operational needs.

To further enhance system performance and reliability, an Energy Management System (EMS) will be deployed. The selected EMS, the eOS platform developed by ELUM, will oversee the integration and coordination of PV generation, diesel generator operation, and load distribution. The EMS includes the ePowerControl module to optimize fuel use and minimize carbon emissions, as well as the ePowerMonitor feature for real-time performance monitoring. This ensures not only technical efficiency but also environmental sustainability in ongoing operations.

Beyond the main camp, the system also addresses the lighting demands of the rig. The design includes the installation of eight portable DC solar-powered tower lights, each equipped with onboard batteries capable of operating for 48 hours on a full charge. These lights are designed with MPPT technology to optimize solar energy capture and conversion. Together, they offer a total generation capacity of 10.26 kW, producing 246.2 kWh per day and approximately 89.9 MWh per year. Complementing this, 32 solar-powered crown lights will be installed on the rig tower, with eight lights on each side to ensure complete site illumination. These lights collectively offer a total generation capacity of 6.68 kW, producing 160.3 kWh daily and 58.5 MWh annually. Their integration into the hybrid system significantly reduces diesel consumption for lighting, while maintaining consistent and reliable performance.

Solar-battery- diesel hybrid system deisgn for the rig drilling operation

Due to the critical importance of the drilling operations, a decision was made to design a solar-battery-diesel hybrid system exclusively. The implementation of this system will be assessed based on the performance of the camp and lighting system project. Drilling operations involve heavy power demands, including

equipment such as mud pumps, top drives, and rotary tables, which represent the standard operational load. However, in the event of a stuck pipe, power requirements can increase by 50% to 70% above the normal load. The rig is equipped with four 1750 kVA (1200 kW) DGs.

Drilling rig operation load

According to the engineering team, normal drilling operations are typically powered by three out of four 1750 kVA (1200 kW) DGs. These generators operate at an average load of approximately 80% of their combined capacity. This results in an average continuous power demand of 2,880 kW, as the drilling operations run 24/7 without interruption.

Stuck pipe; the worst case

the worst-case scenario, such as a stuck pipe situation, the power demand can surge to 4 MW to free the stuck pipe. During these instances, all four 1750 kVA DGs are required to operate simultaneously to meet the increased load and successfully perform the operation.

System design based on the operation data

The information previously gathered has been applied to develop a tailored solar-battery-diesel hybrid system designed to support the rig's drilling operations. This system is engineered to accommodate both standard operational demands and the elevated power requirements that arise during exceptional events, such as a stuck pipe scenario. In such instances, the system is configured to utilize battery power to handle the initial minute of the high-demand load, after which diesel generators assume full operational control to sustain the load for the remainder of the event.

Design considerations were strongly influenced by site-specific environmental conditions. Eastern Saudi Arabia experiences high levels of solar irradiation, typically ranging from 6.0 to 6.5 kWh/m²/day, which translates into six to eight hours of effective solar generation each day. However, ambient temperatures can reach extreme levels between 35°C and 50°C, necessitating the use of durable and thermally stable components. Accordingly, the design integrates monocrystalline photovoltaic panels for their efficiency and temperature resilience, as well as lithium iron phosphate (LiFePO₄) battery systems with active cooling to maintain reliable performance under harsh climatic conditions [4].

The system has been strategically sized to optimize solar energy utilization throughout daylight hours, thereby minimizing reliance on diesel fuel. The solar power array is expected to contribute approximately 40% of the rig's normal operational load, resulting in an installed capacity of 1.2 MWp. This solar setup will significantly reduce diesel consumption during daylight operations. The PV panels will be mounted on fixed-tilt structures oriented according to the site's latitude to ensure stable generation throughout the year. High-temperature-tolerant inverters will be employed to manage energy conversion efficiently and mitigate power losses.

To complement the solar infrastructure, a battery energy storage system (BESS) is incorporated into the design. The BESS will play a critical role during high-load anomalies, particularly stuck pipe situations, by delivering 4 MW of power for the first minute of the event. This necessitates a minimum battery capacity of approximately 87 kWh, inclusive of a safety margin. Beyond peak-load support, the system will store surplus solar energy during the day, with an additional battery capacity ranging between 500 and 1,000 kWh allocated for continuous support of normal drilling activities. LiFePO₄ batteries are selected for their thermal stability and long lifecycle, and the battery system will include a robust battery management system (BMS) along with active cooling components to safeguard performance and safety.

The existing diesel generators remain a fundamental part of the hybrid configuration. Under standard conditions, three of the four available generators—each rated at 1,750 kVA (1,200 kW)—operate at around 80% capacity to deliver a continuous power output of approximately 2,880 kW. The fourth generator is kept on standby to ensure system redundancy. During high-demand scenarios, such as those induced by drilling

complications, all four generators are activated to supply the full 4 MW load once the battery discharge period concludes. To enhance reliability and extend equipment lifespan, generator usage will be rotated periodically to balance operational wear. Moreover, enhanced cooling systems will be introduced to protect generator performance against the region’s extreme temperatures. The hybrid setup will be governed by an advanced energy management system (EMS), ensuring synchronized operation among all components to optimize fuel efficiency and reduce environmental impact.

The following table summarizes the final system configuration developed for the solar-battery-diesel hybrid system supporting the rig’s drilling operations.

Table 2—A summarized system configuration for the hybrid system intended for the drilling operation

Component	Specification/Capacity
Solar Power System	~1.2 MWp (monocrystalline panels, temperature-tolerant)
Battery Storage (BESS)	~1 MWh (includes ~87 kWh for stuck pipe scenario)
DGs	4 × 1750 kVA (1200 kW each)
Hybrid Controller	Advanced EMS with temperature management capabilities

Forecasted results from the design

The forecasted savings for the solar-diesel hybrid system and the planned solar-battery-diesel hybrid system highlight significant potential for reducing diesel consumption, fuel costs, and CO2 emissions. This section focuses on the expected savings from both systems, demonstrating their impact on operational efficiency and sustainability.

Solar-Diesel Hybrid System for Camp and Lighting

The implementation of the solar-diesel hybrid system for the rig’s camp and lighting has resulted in substantial energy and environmental benefits by seamlessly incorporating solar power into the site’s daily operations. Annual diesel consumption is projected to decrease by approximately 207,676 liters across the camp, tower lights, and crown lights. This reduction in fuel usage translates directly into cost savings, with an estimated annual financial benefit of 63,691 SAR, or roughly 17,050 USD, based on the diesel price of 1.15 SAR per liter. In addition to the economic impact, the system contributes significantly to environmental sustainability. Given that each liter of diesel produces approximately 2.68 kilograms of CO₂, the reduced fuel consumption corresponds to a projected annual decrease of 557 metric tons of carbon dioxide emissions. These results underscore the hybrid system’s effectiveness in enhancing operational efficiency while aligning with broader environmental and sustainability objectives.

Solar-Battery-Diesel Hybrid System for Drilling Operations

The planned solar-battery-diesel hybrid system for drilling operations is specifically engineered to accommodate the rig’s substantial power requirements, particularly during high-load scenarios such as stuck pipe events. By partially offsetting the energy demands of critical drilling equipment, including mud pumps, top drives, and rotary tables, the system is projected to reduce diesel consumption by approximately 40%. This reduction corresponds to annual fuel savings of around 1.2 million liters. At the prevailing diesel price of 1.15 SAR per liter, the anticipated cost savings amount to approximately 1.38 million SAR, equivalent to roughly 368,000 USD per year. In addition to the economic advantages, the environmental impact is also significant. With each liter of diesel generating 2.68 kilograms of CO₂, the system is expected to prevent the release of approximately 3,180 metric tons of carbon dioxide annually. These projected savings demonstrate the system’s potential to meaningfully reduce both operational costs and environmental footprint, reinforcing the value of hybrid energy solutions in high-demand oil and gas applications.

Implementation of solar-diesel hybrid system for the rig camp and lighting

The implementation of the Solar-Diesel hybrid system at the candidate rig followed a phased and adaptive deployment strategy. For the main camp, a single portable solar generator was installed, delivering a total capacity of 70 kW. This represented a scaled-down version of the original design, which had proposed the deployment of three generators with a combined output of 210 kW. As a result of this adjustment, the number of required DC/AC inverters was also reduced, from four units to one.

In a similar fashion, photovoltaic module installation on the caravans was limited during the initial implementation. Instead of equipping all forty caravans with rooftop PV modules as originally intended, only a single caravan was fitted with solar panels, providing a capacity of 6 kW rather than the projected 180 kW. Consequently, the number of inverters needed for this configuration was reduced from three to one.

For the lighting systems, the full-scale design was executed as planned. Eight DC solar-powered portable tower lights were successfully installed, fully replacing the previously used diesel-powered units. In addition, the crown lighting system was upgraded to solar power in its entirety. A total of thirty-two solar-powered lights were mounted at the top of the rig tower, with eight lights positioned on each side to ensure complete coverage. These installations replaced the existing diesel-powered crown lights, completing the transition of the rig's lighting infrastructure to renewable energy.

Major findings

This section summarizes the major findings and results arising from the implementation of the solar-diesel hybrid system at the rig camp and lighting. The outcomes demonstrate the technical feasibility, environmental value, and cost-effectiveness of integrating renewable energy solutions into conventional oil and gas operations.

The main camp

The integration of solar power into the main camp's energy infrastructure resulted in a measurable reduction in diesel consumption, amounting to approximately 56,700 liters annually per rig. This outcome highlights the potential for significant fuel savings when solar capacity is effectively incorporated into non-critical rig systems. When extrapolated across a fleet of 50 rigs, the total annual diesel savings could reach 2.84 million liters, representing a major step toward decreasing dependence on fossil fuels. In parallel, the environmental benefits are equally substantial. Using the standard emissions factor of 2.68 kg CO₂ per liter of diesel, this translates to a reduction of 152 metric tons of CO₂ emissions per rig per year. Multiplied across 50 rigs, the total avoided emissions exceed 7,600 metric tons of CO₂ equivalent annually, supporting national and corporate emissions reduction targets. Economically, the diesel savings correspond to an annual cost reduction of approximately 65,205 SAR per rig, or 3.3 million SAR for the entire fleet. These figures confirm the strong financial case for renewable integration in rig camp operations, offering both environmental and economic value with minimal operational disruption.

The portable tower lights

The implementation of solar-powered tower lighting represents a complete transformation of the rig's lighting system from fossil-fuel dependence to 100% renewable energy. Each of the eight installed tower lights is equipped with a solar array and integrated battery backup, providing up to three days of autonomous operation without requiring diesel input. This solution not only eliminates diesel usage for lighting but also removes the need for generator support or fuel logistics for this segment. The resulting annual fuel savings of 100,000 liters per rig is substantial, especially considering the relatively modest power demand of lighting systems. Furthermore, the replacement of diesel-powered tower lights improves site safety and reduces noise and air pollution, all while ensuring continuous and reliable illumination. The initiative sets

a replicable model for other rigs seeking to modernize lighting infrastructure without compromising on performance or reliability.

The crown lights

The upgrade of the rig's crown lighting to solar power complements the transition already seen with the tower lights. A total of 32 solar-powered crown lights were installed—eight on each side of the rig tower—to replace the diesel-dependent system previously in use. Although detailed diesel consumption data specific to this component was not separately itemized, its integration into the overall solar-diesel hybrid system further contributes to fuel and emissions reductions. From a functional standpoint, the new crown lights ensure uninterrupted visibility at height, a critical requirement for safe rig operations. The project underscores the practicality of using solar energy in elevated and exposed rig structures, where wiring and maintenance for diesel systems are both costly and complex. In conjunction with other solar initiatives, the crown lighting upgrade supports the broader objective of building low-emission, high-efficiency rig environments.

Conclusion

The implementation of the Solar-Diesel hybrid system at the candidate rig represents a significant step toward reducing reliance on diesel fuel in off-grid O&G rigs. By integrating solar power into the camp and lighting systems, the project has demonstrated measurable benefits in reducing diesel consumption, lowering CO₂ emissions, and minimizing operational costs. These outcomes underscore the feasibility of renewable energy solutions in achieving sustainable development goals.

The solar-powered camp and lighting systems achieved considerable reductions in fuel consumption and carbon emissions, paving the way for enhanced operational efficiency and environmental stewardship. Additionally, the transition to solar-powered portable and crown lights resulted in a 100% reduction in diesel reliance for lighting systems, showcasing the potential of renewable energy to transform energy-intensive operations.

The success of this initiative lays a strong foundation for further adoption of hybrid systems across additional rigs. The planned solar-battery-diesel hybrid system for drilling operations is expected to build on these results, delivering even greater energy savings and environmental benefits.

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